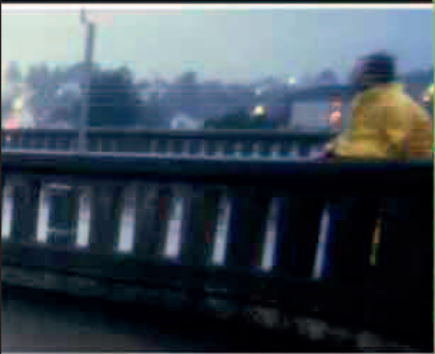




Rod-shaped, helical bacteria, known as *Helicobacter pylori*, attached to the stomach lining. These bacteria cause stomach ulcers.

“ I DRANK IT DOWN IN ONE GO AND THEN **FASTED** FOR THE REST OF THE DAY. A FEW **STOMACH GURGLES** OCCURRED, **WAS IT THE BACTERIA** OR WAS I **JUST HUNGRY?** ”

Barry Marshall, Australian physician, from Nobel Lecture, December 8, 2005



Full orchestra

A MIDI controller keyboard came with a standard set of MIDI instruments, but MIDI messages can play any sounds.

openings connect to major arteries and veins

polyester shell



THIS YEAR, THREE IMPORTANT DEVELOPMENTS

in consumer electronics reflected advances made exactly 10 years earlier. The first day of the year was the deadline for all computers connected to the worldwide network known as ARPANET to be using the internet protocol, TCP/IP (see 1973). Many computer historians consider that day as the **switching on of the modern internet**. Until then, some computers were still using a different protocol to communicate, but TCP/IP has formed the **basis of all internet traffic ever since**. The first computer to have a graphical user interface (GUI) was the Alto, developed at Xerox's research center in California (see 1973). Ten years later, Apple Computer Inc. launched the **first personal computer with a GUI**.

In October, American businessman David Meilahn made the first commercial mobile phone call, over a cell-based wireless network—10 years after the first call on a prototype device (see 1973).



Apple Lisa

The Apple Lisa, the first personal computer with a graphical user interface, and the team that designed it.

Physicists at the **European Organization for Nuclear Research (CERN)** discovered **three particles** whose existence gave more weight to the Standard Model (see 1974). The W^+ , W^- , and Z bosons carry the "weak interaction," which is involved in radioactive decay (see pp.266–67). The existence of these particles had been **predicted in 1968**, as a result of a unified theory of the weak interaction and the electromagnetic force. Their discovery was only possible because of CERN's powerful new particle accelerator, the **Super Proton Synchrotron**, which had been operational since 1976.

In October, the 17th General Conference on Weights and

Measures defined the meter as equal to the **length of the path traveled by light in a vacuum** during a time interval of one 299,792,458ths of a second.

At the University of Western Australia, in Perth, physician **Barry Marshall** (b.1951) and pathologist **Robin Warren** (b.1937) identified the most **common cause of stomach ulcers**—a common condition that **can lead to death** by gastrointestinal bleeding or stomach cancer. Their work began when Warren became curious about large numbers of a new species of bacteria he had found in patients' stomachs. At the time, no one expected bacteria to be present inside the stomach, because of the strong acid there. After finding the bacteria in patients with stomach ulcers, Warren and Marshall hypothesized that these organisms were infecting the lining of the

stomach and the duodenum (part of the small intestine), leading to inflammation and causing the ulcers. Marshall realized he needed to carry out a definitive **test of the hypothesis on a human subject**—and he decided to use himself as the subject. After making sure the bacteria were not present in his stomach, he prepared a culture, mixed it with chicken broth, and drank it. Marshall's stomach lining became inflamed, and the hypothesis was soon proven. The link between the new species of bacteria—later named *Helicobacter pylori*—and stomach ulcers meant that most ulcers can be treated using antibiotics, saving hundreds of thousands of lives. In recognition of their discovery, Marshall and Warren were awarded the 2005 Nobel Prize in Physiology or Medicine.



Mobile phone

The world's first commercially available mobile phone, Motorola's DynaTAC 8000X. Early mobile phones like this were affectionately known as "bricks."

7.5 THE NUMBER OF **TIMES LIGHT** CAN **TRAVEL** AROUND **EARTH** IN A **SECOND**

October Introduction of MIDI (Musical Instrument Digital Interface)

December 2 American Barney Clark becomes the first person to be given a **permanent artificial heart**

January 1 The date on which the modern internet is "turned on"

January Particle physicists at CERN discover the first particles **responsible for the weak nuclear force**, the two W bosons

January 19 Apple releases its Lisa computer, the first home computer with a graphical user interface (GUI)

June First description of a new species of bacteria, *Helicobacter pylori*, which survives in the stomach and causes stomach ulcers

May Particle physicists at CERN discover the other particle responsible for the weak nuclear force, the Z boson

October 21 The meter defined as equal to the length of the path traveled by light in a vacuum during a set time interval

October 13 First wireless call made, on a Motorola DynaTAC 8000X, the world's first commercial portable mobile phone